Amateur Aerial Surveying



Jul 16, 2011
State of the Map EU

Andrzej Zaborowski @ Intel

Imagery is useful

Overhead imagery is useful

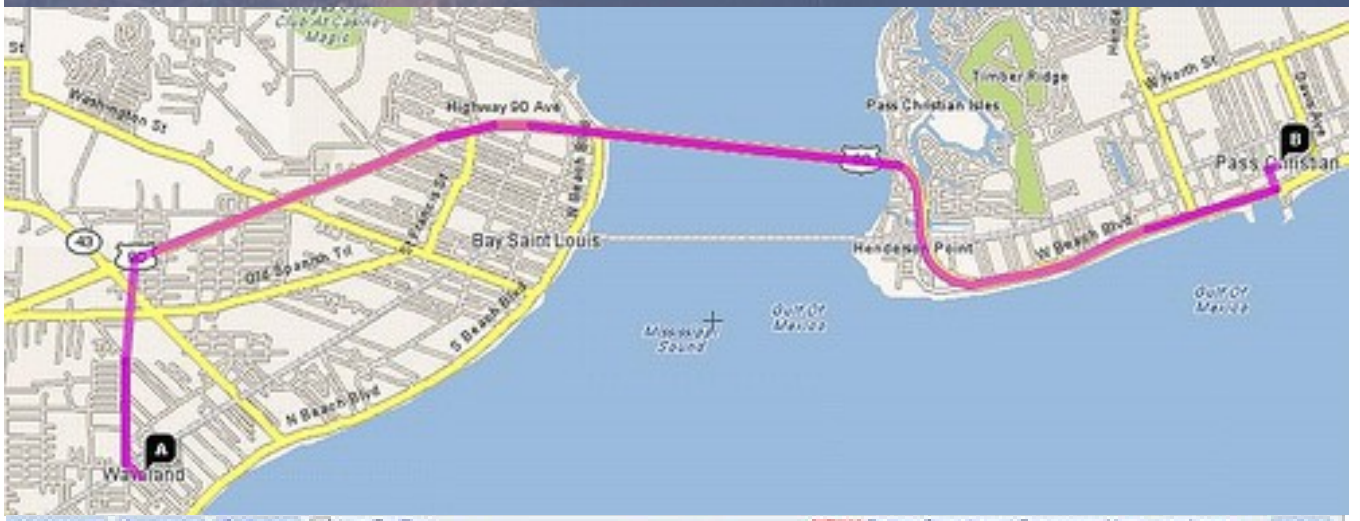
- adding coastlines with GPS?
- reference for OSM
- pretty background for the map
- non-OSM uses, automatic processing

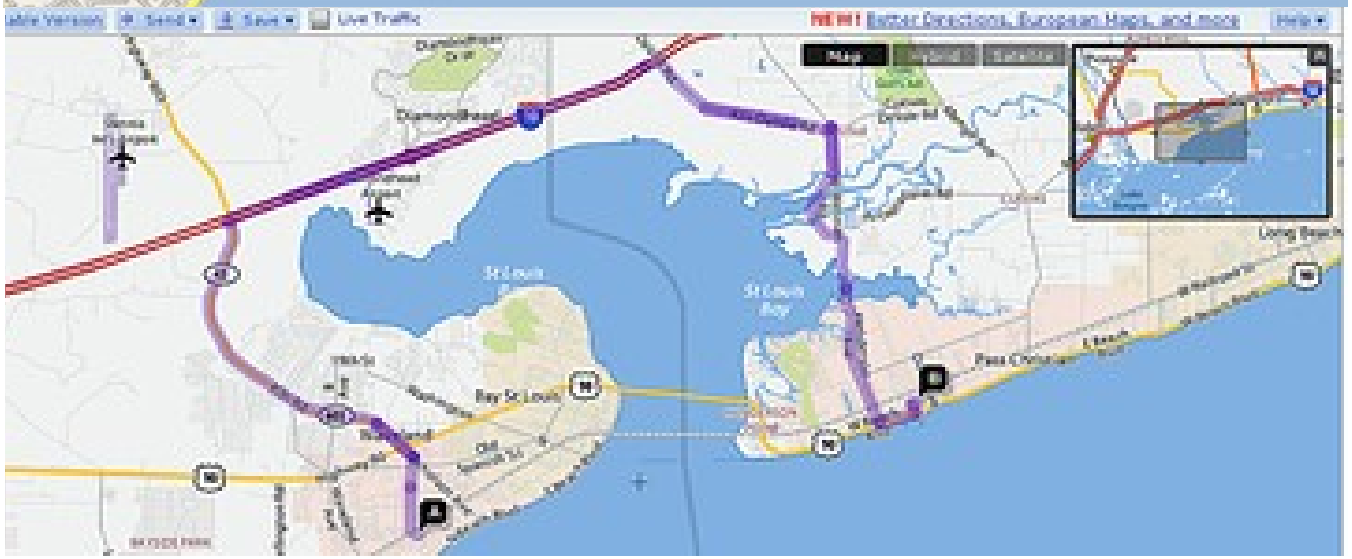
Imagery is useful

- poor availability outside select areas, expensive
- usually out of date
- outside of US non-free – not free enough for OAM anyway, Ms, Y!: custom license, Space Chapter NC
- Ms: $20/km², very long term
- satellite imagery strictly controlled, lower resolution, no oblique images

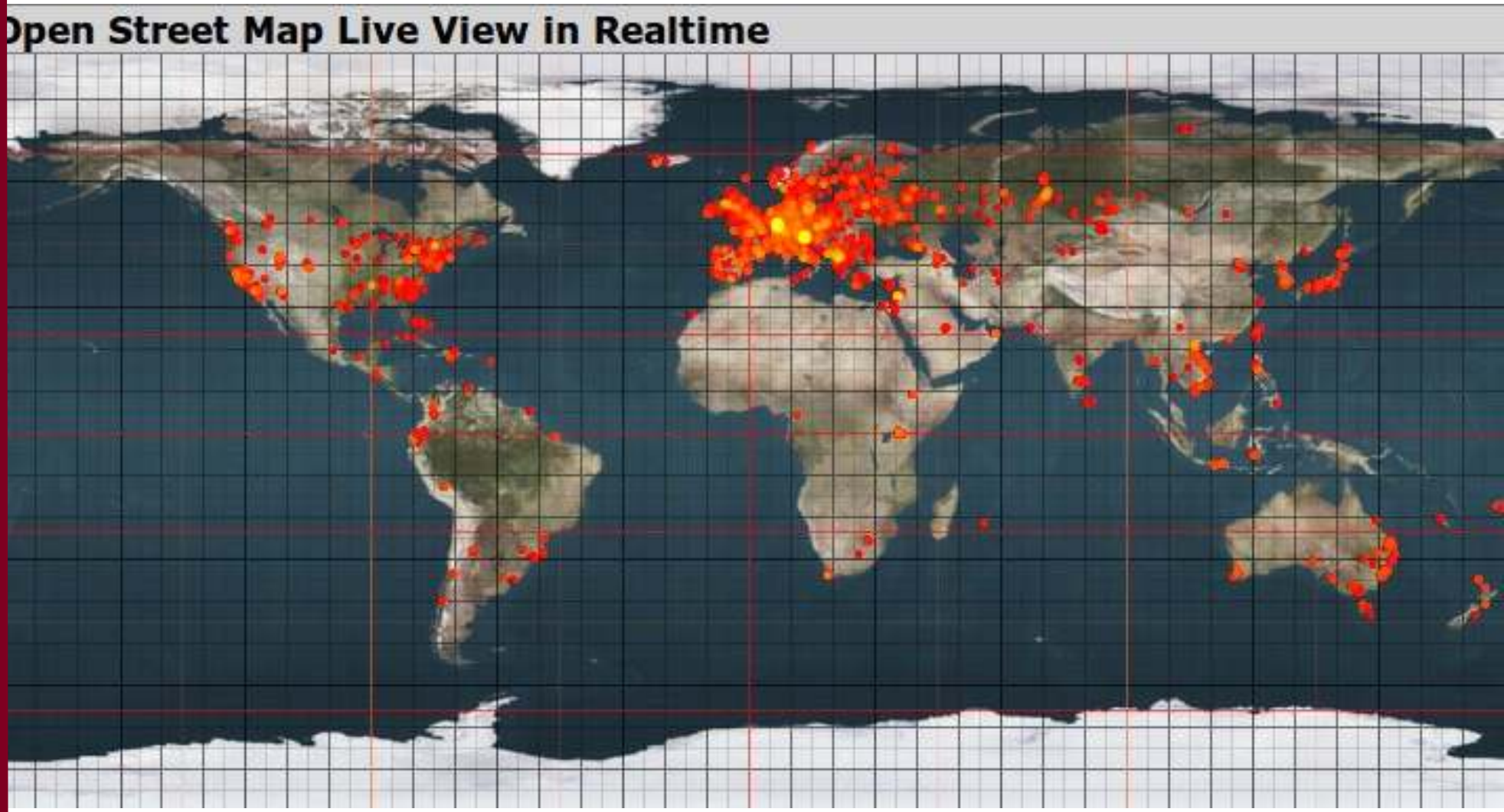
Imagery is useful







Would be cool to crowdsrc



Would be cool to crowdsrc

- always current when needed
- not directly business-decided coverage
- free to upload to a single global repository like OAM (soon to be)
- community building
- available ahead of a disaster

What methods are there?

- Traditional: manned airships, Now: unmanned
- kites
- drones: RC copters, planes
- balloons, blimps
- many more: slingshot cameras, water rockets, nanosats, small sats







There are more methods



75g, 1903!

Kites, quick facts

- costs: < $300 a basic kit
- altitudes: avg 400-500m, in good conditions 800-900m, record 8+km (FL250+)
- no horizontal control
- takes about 1h to set up, fly and pack
- “kit” can be as little as 1kg, easily fits in a backpack
- weather conditions some 1-2 days in a week (bp figure) avg for Poland

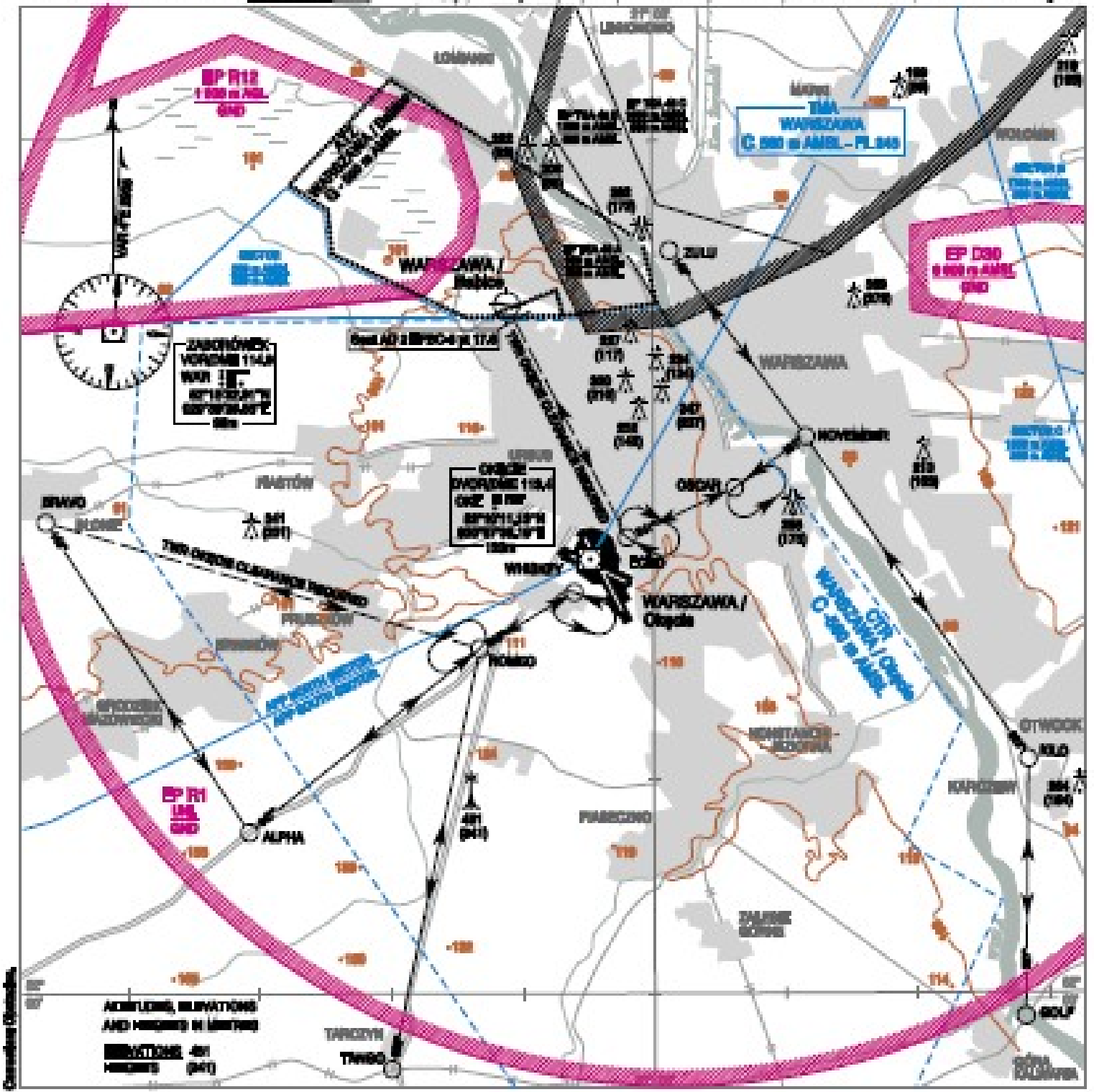
Kites, how to start

- wiki.grassrootsmapping.org!
- kites and line in EU: kapshop.nl or local makers
- ALWAYS WEAR GLOVES
- know local airspace law, not completely uniform across EU

AERODROME ELEV 110 m
HEIGHTS RELATED TO AD ELEV

VFR
ARRIVAL, DEPARTURE AND TRANSIT ROUTES

WARSZAWA / Okęcie



Kites, problems

- legal
- technical
- screw-ups





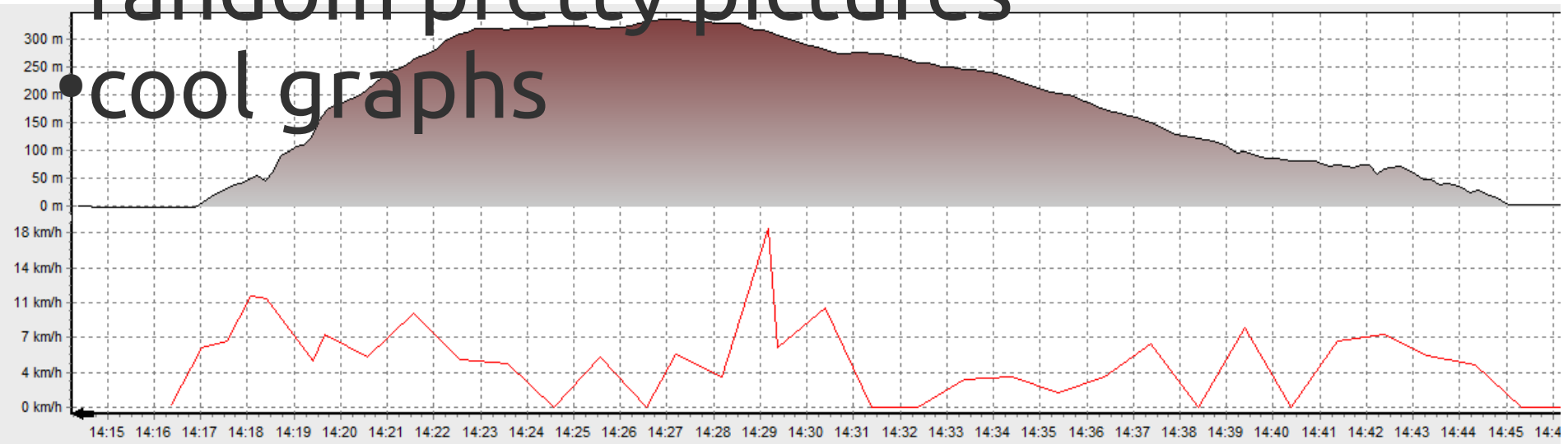
Kites, sustainable?

- lots of talk
- cost per km² handwaving
  - area covered: avg 5km² max 15km² at Yahoo/Bing resolution or better, ignoring the tilt
  - 1-2 people necessary
  - 45-60min
- what are the results
- sustainable? IMO yes, or close to be

Some 50 locations

Kites, added value

- kites are fun!
- spur people's interests
  - mapping with the scouts
  - a good time to give out leaflets
- random pretty pictures
- cool graphs



Name 01.11/14:14

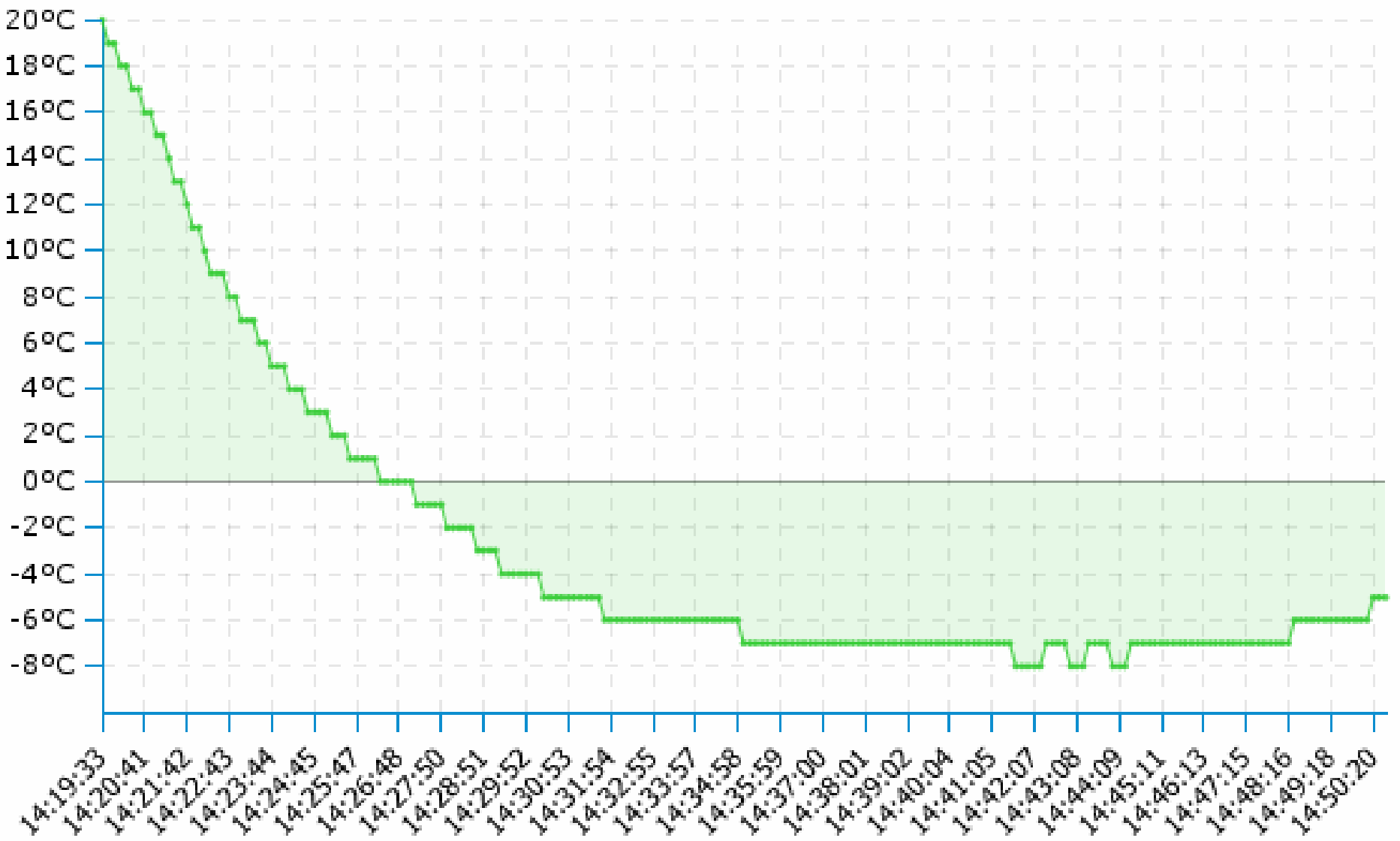
Notes

Preview

Start time	2010-11-01 14:14:23	Highest point	337 m
End time	2010-11-01 14:47:16	Lowest point	-3 m
Activity time	00:32.53	Ascent	365 m
Activity distance	0,60 km	Descent	363 m
Paused time	00:00.00	Max speed	22 km/h
Paused distance	0,00 km	Avg speed	2 km/h
Total time	00:32.53	Trackpoints	35
Total distance	0,60 km	Laps	1

?

Camera sensor temperature



H

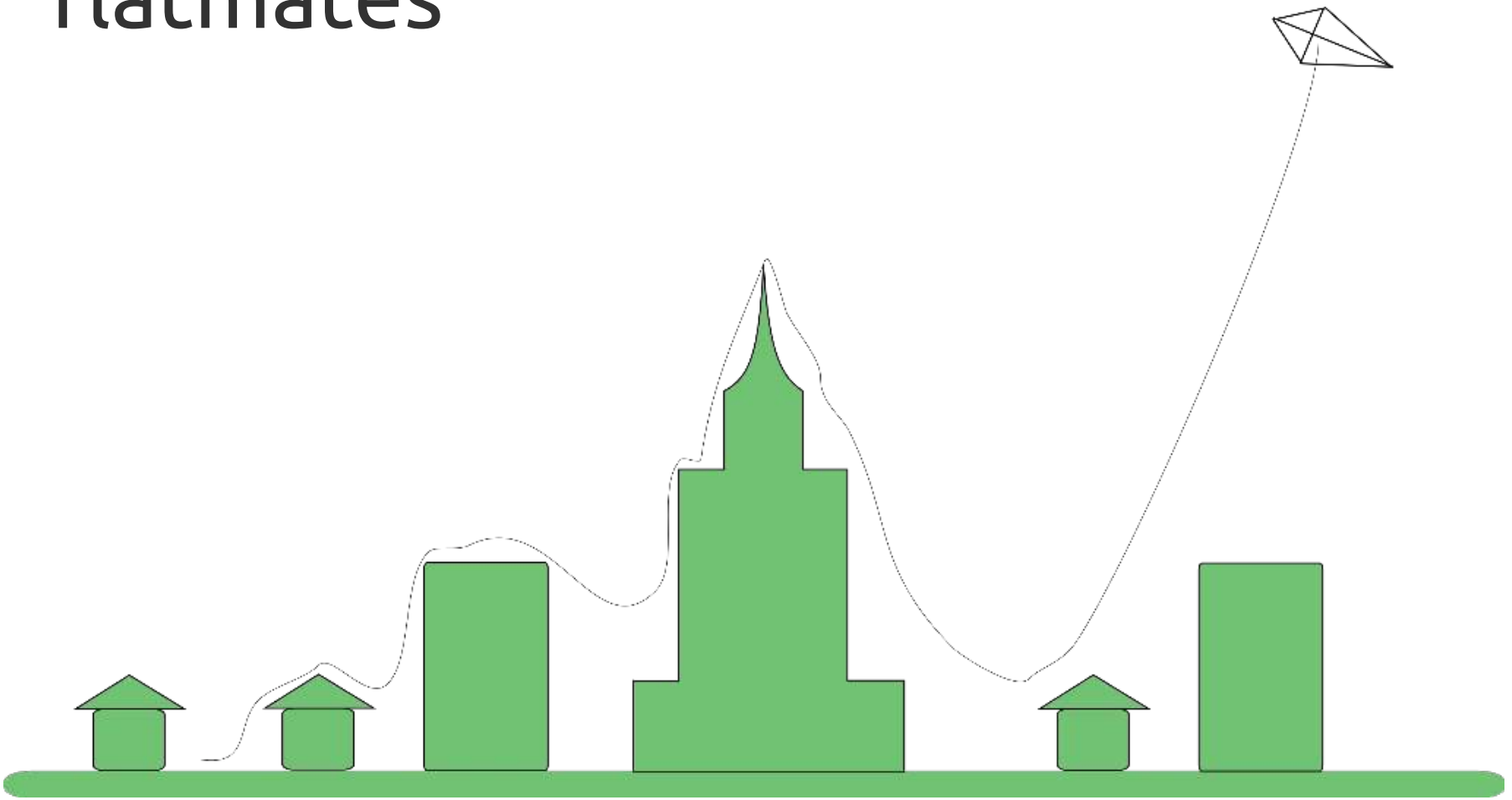


Kites, added value

- GrassrootsMapping & Public Laboratory
  - IR images, NRG & NDVI (sp?)
  - social impact
  - documentation
  - use Google Maps imagery as reference for most things :-(

Other funny aircraft

- Multicopters, planes, nano satellites, drag kites, water rockets and more.
- Hydrogen & CNT: problem with flatmates



Processing, rectification

- Currently quite hard using open source
- “brute force” method: a lot of GCPs
- correct projective mapping with GCPs:
  - Cartagen Knitter: maps.cartagen.org, try it!
  - GrassGIS
  - ILWIS
  - Since recently: qGIS
- Intelligent:
  - Hugin: wiki.openstreetmap.org/wiki/User:Balrog/Aerial_Imagery/Rectification
  - pix4d.com and friends

http://wrp.geothings.net/mapscans/68



Canning Town | warp it

"Canning Town" by elbisreverri on flickr CC-BY-NC-SA http://flickr.com/photos/elbisreverri/445202836/

73 gcps. Added about 1 month ago

WMS Original Download Warped Metadata





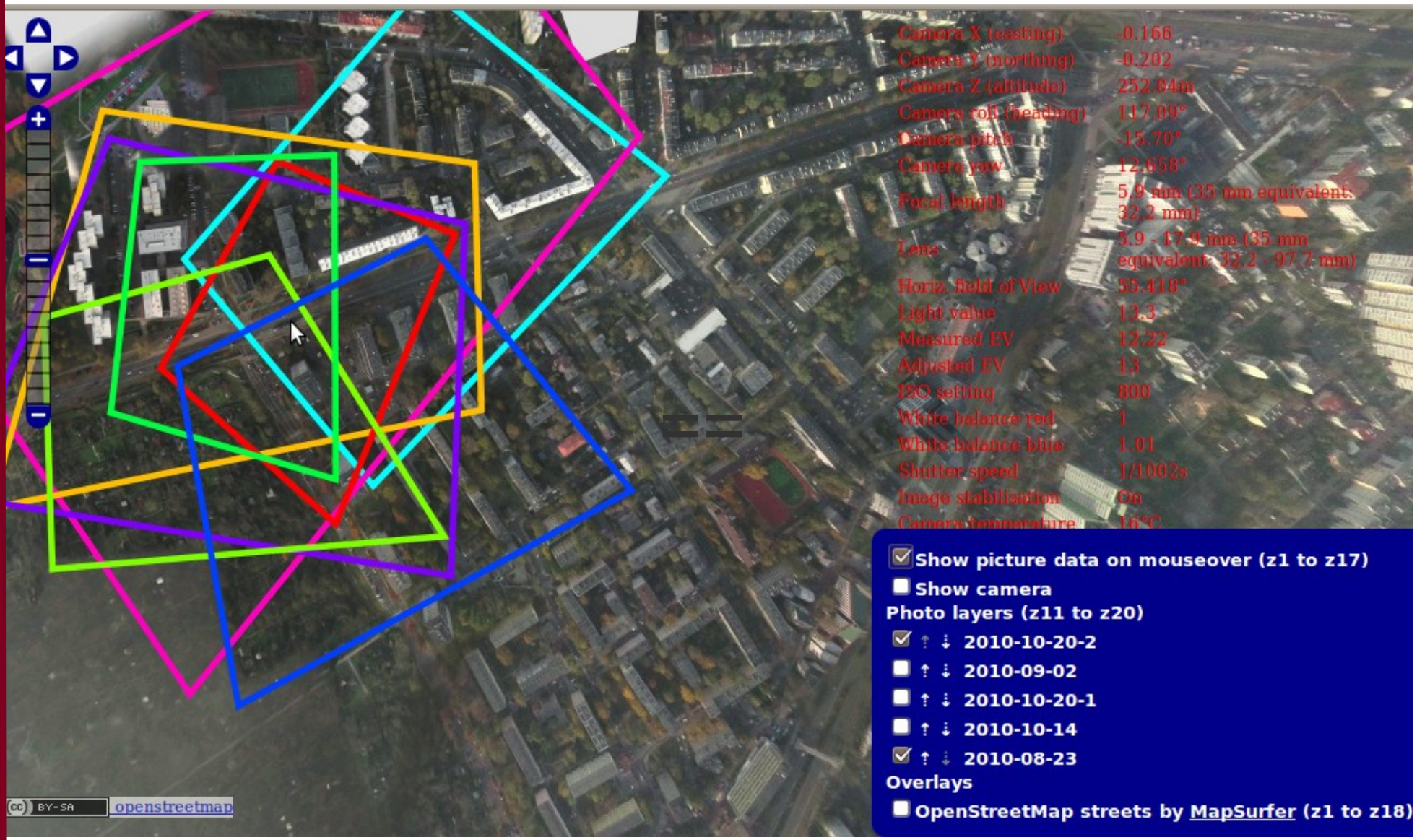
Double-Click on the scanned map, and the reference map, then click **Add Control Point** *. Do this for at least 3 points.*

Ground Control Points

1	788	256	0.016597509	51.51700282	Delete
2	520	224	0.013443231	51.51814446	Delete
3	535	719	0.011855363	51.51343082	Delete
4	925	606	0.015653371	51.51356436	Delete
5	974	426	0.017327070	51.51484630	Delete
6	895	338	0.017219781	51.51586114	Delete

Processing results so far

- http://ooc.openstreetmap.pl/kap-warsaw.xhtml



Processing, other uses

- SLAM: Simultaneous Location and Mapping
- 3D models extraction: Ms Bundler and the University of Washington work, go to Vladimir Elistratov's talk later today (16:20?), also pix4d.com
- Crowdsourced word-wide point cloud??

Processing, other uses

- Crowdsourced word-wide point cloud
  - “competition” isn't there yet
  - automatically gives us: orthoimagery, bird eye, street view, trailview, streetside view, whatever view, OpenDEM and more
  - openspace routing or aid in OSM-based routing
  - building outlines can be automatically extracted, input to OSM

Thank you

